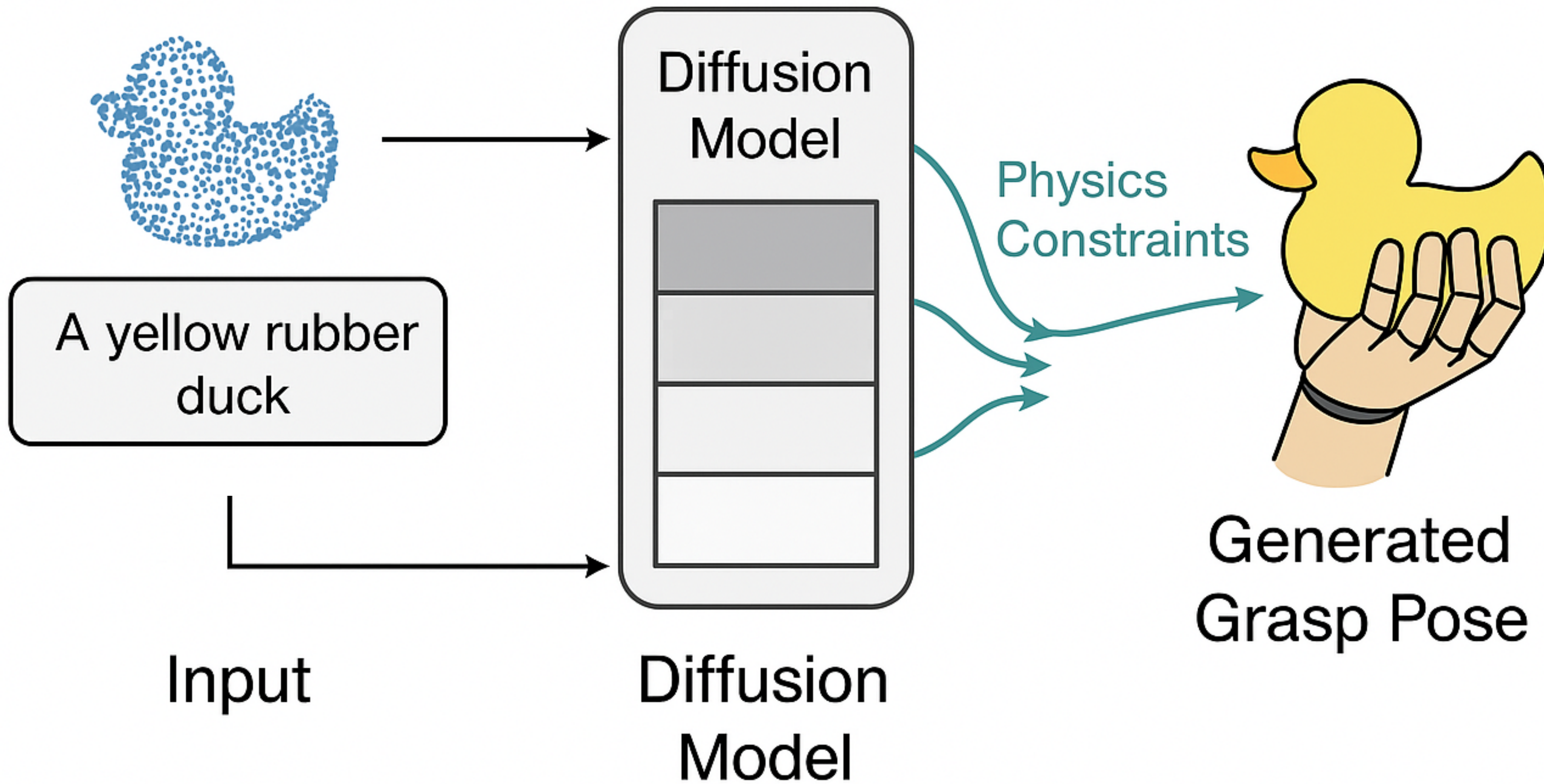


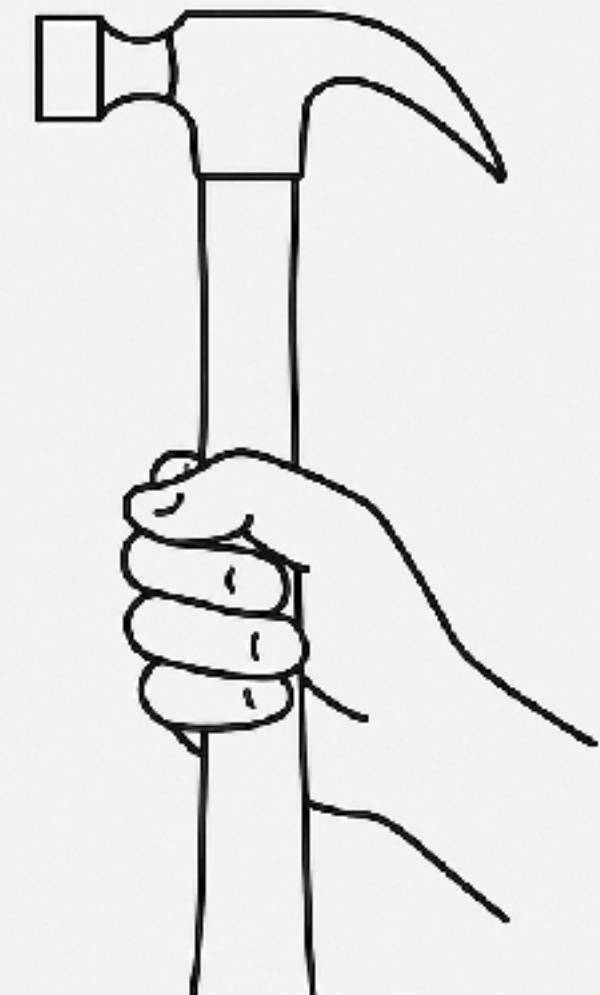
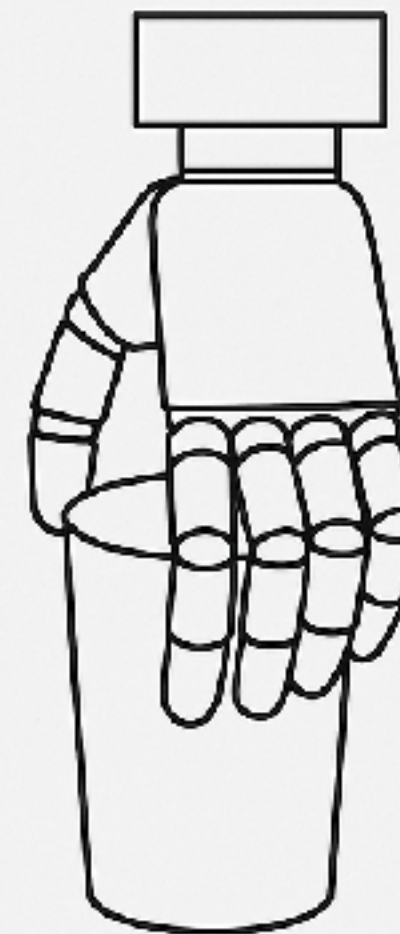
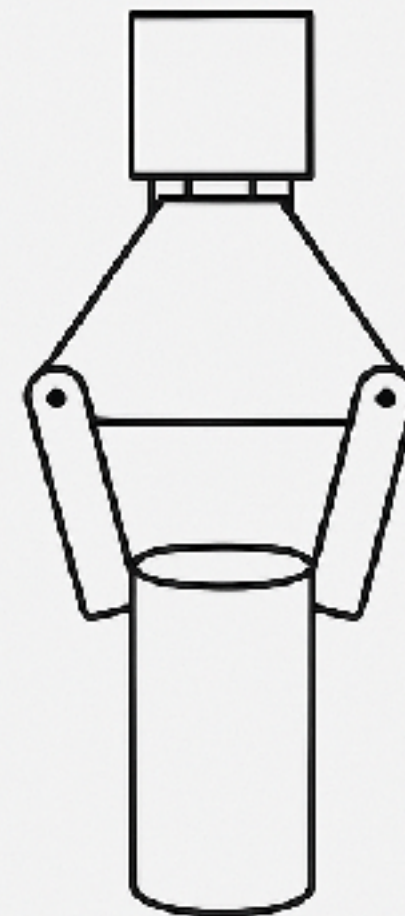
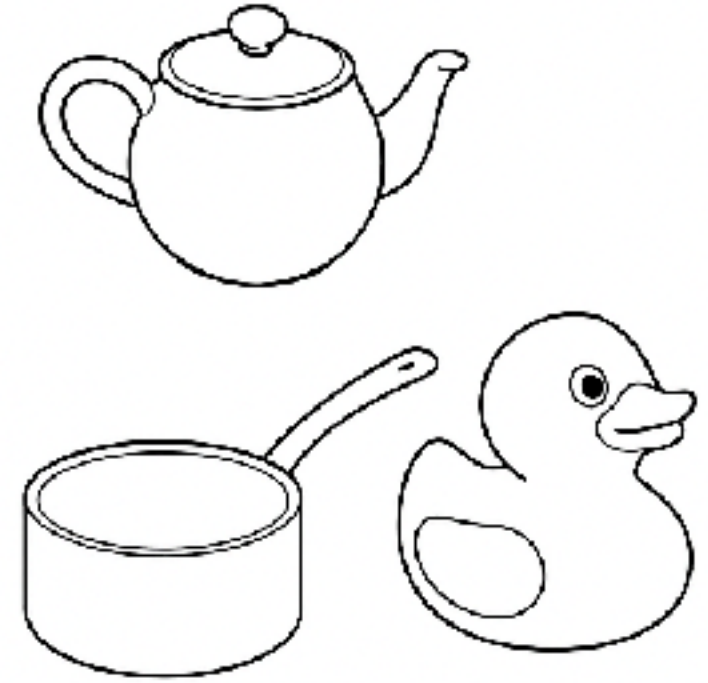
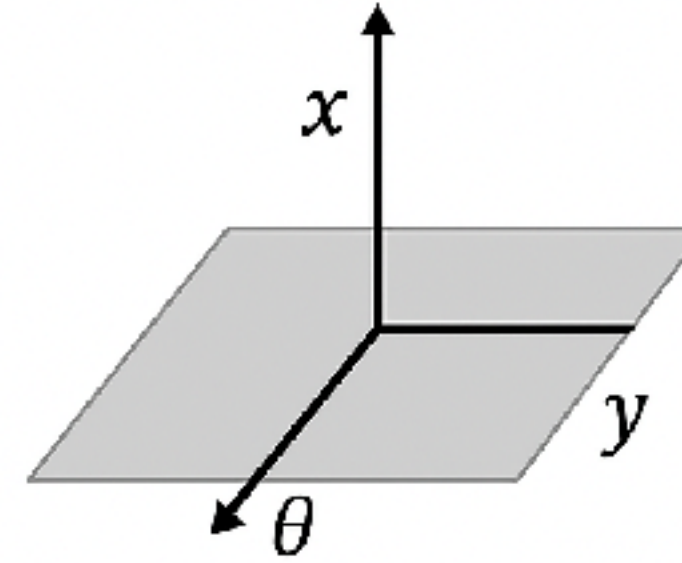
# DexGrasp: Towards Universal Robotic Dexterous Grasping with Physics Awareness





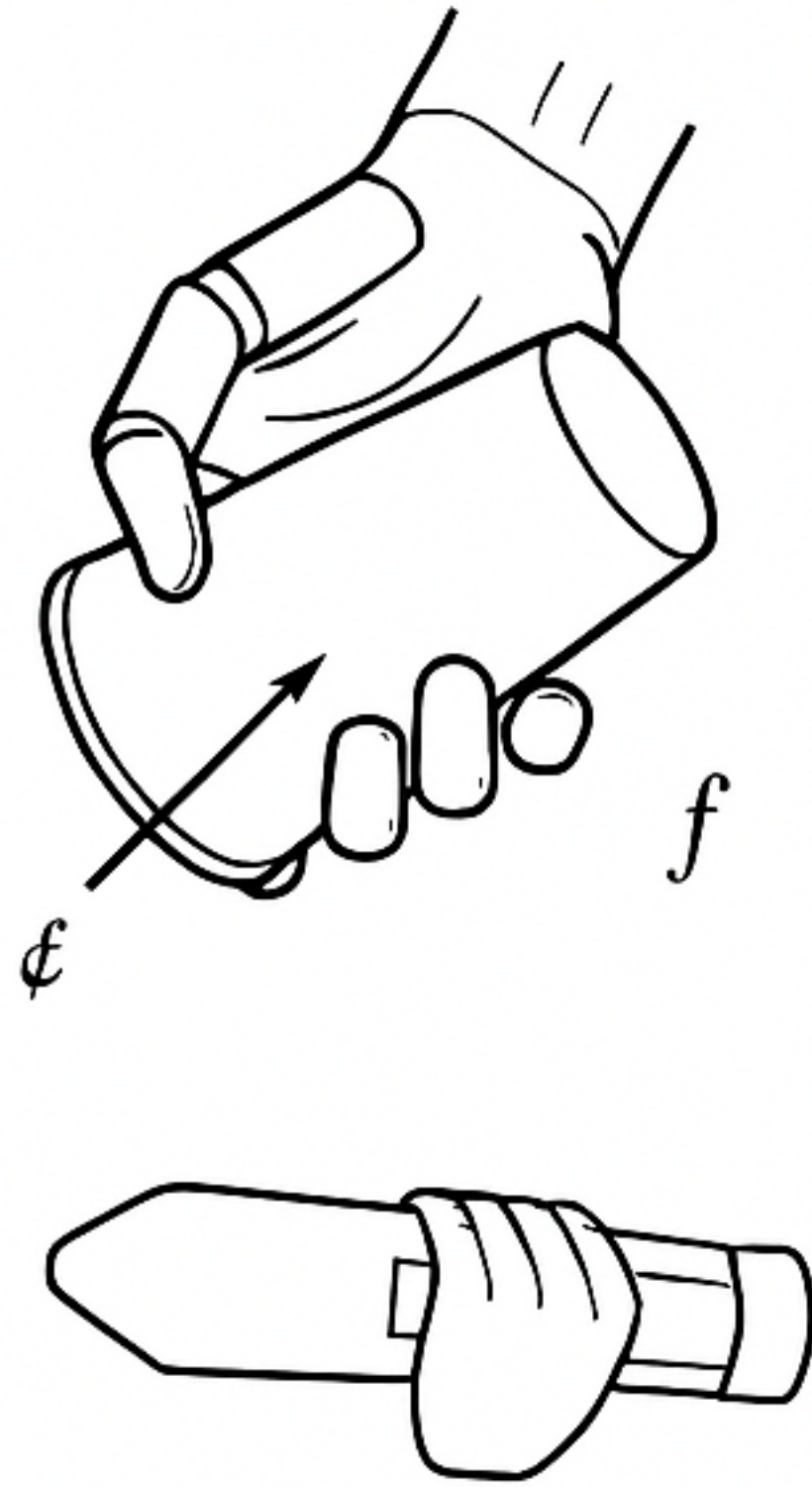
# Real-World Motivation

## Dexterous Grasping Challenges



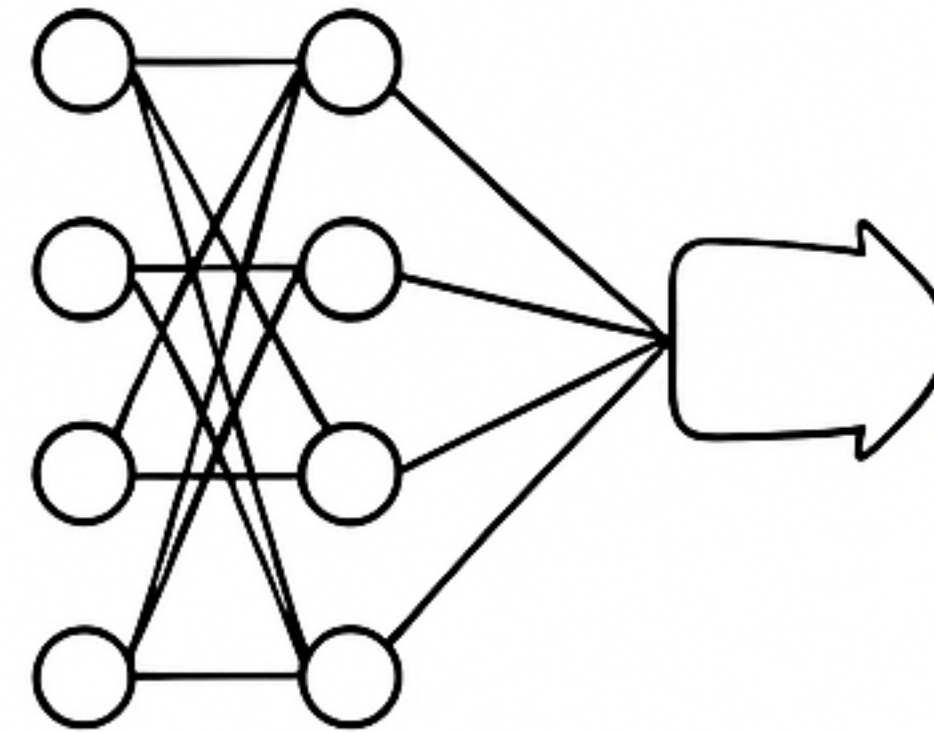


# Three types of crawling strategies



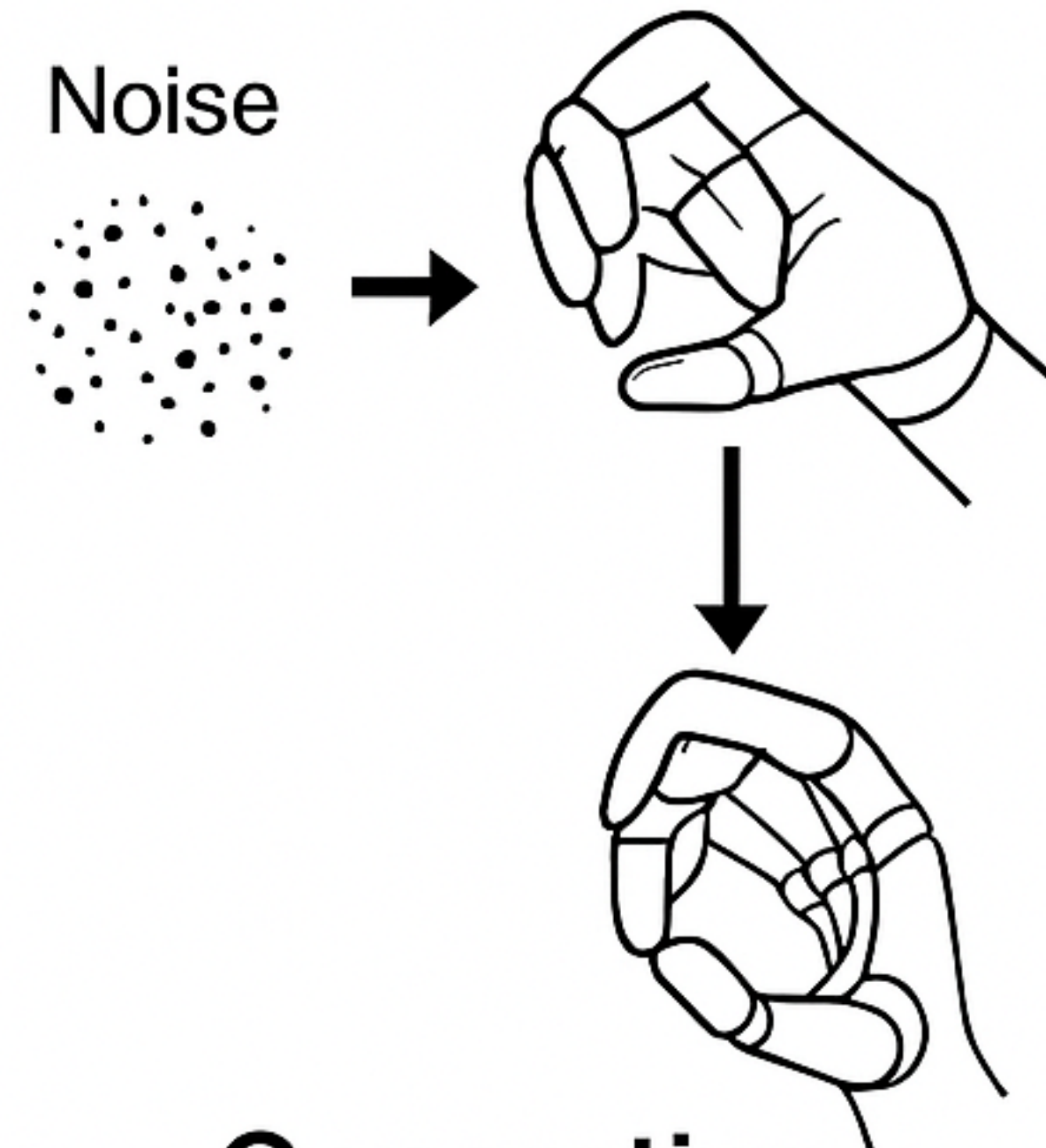
## Rule-based

- ✓ Explainability, controllability
- ⚠ Slow, hard to generalize



## Learning-based

- ✓ Fast prediction, easy deployment
- ⚠ Averaging, local minima



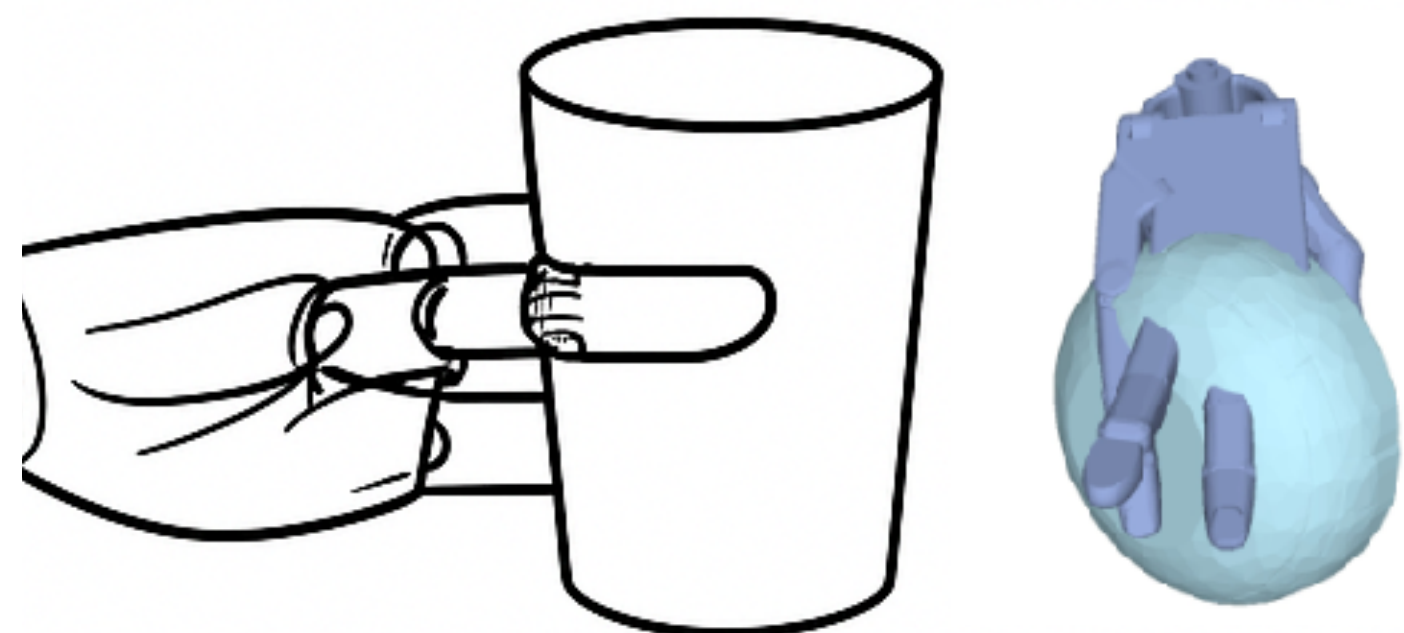
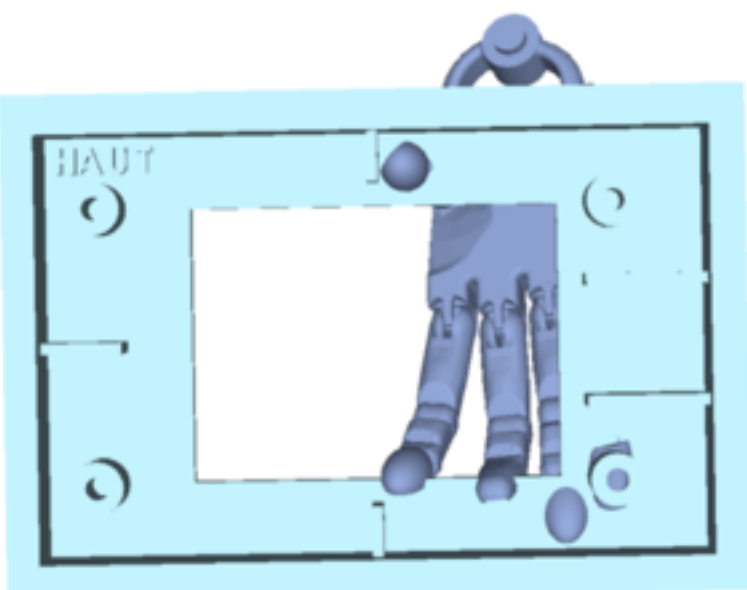
## Generative

- ✓ Diversity, complex distribution
- ⚠ Physics-unaware, unstable

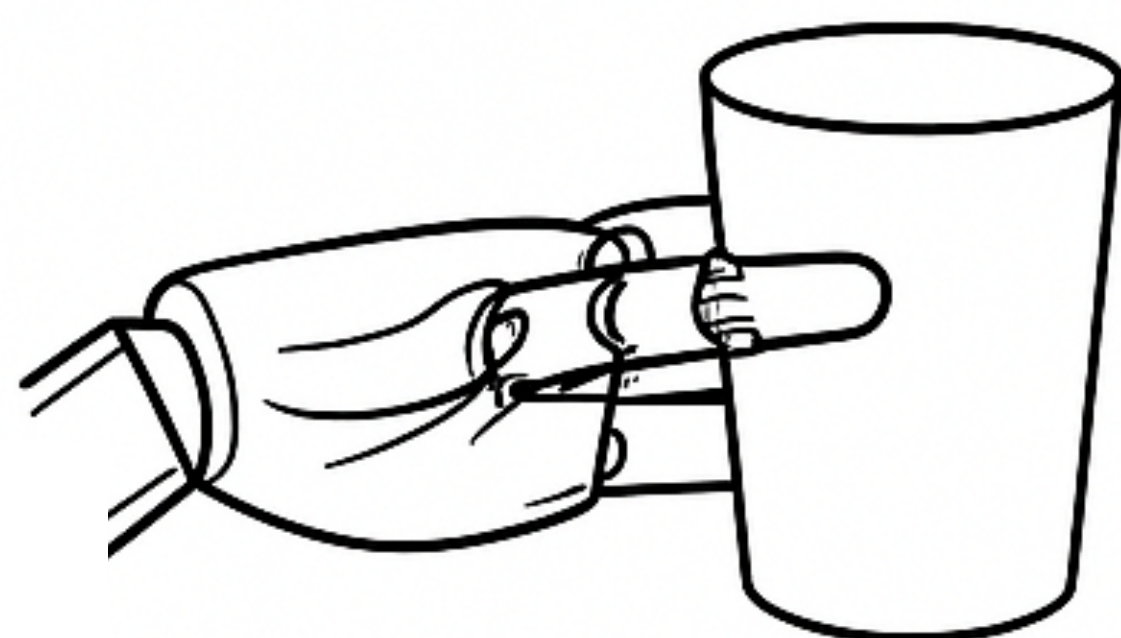


# Why grasping fails – and how physics can help

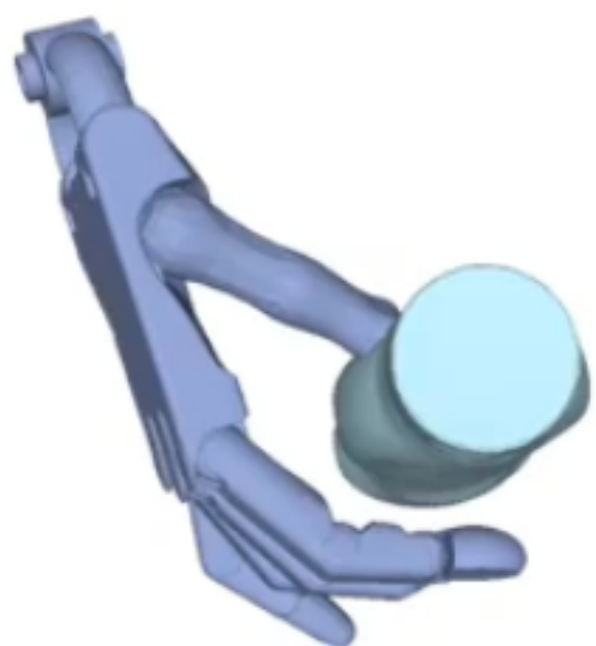
**X** No physical awareness



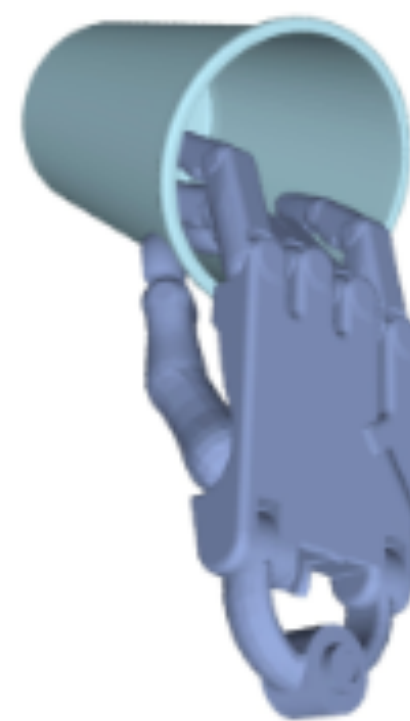
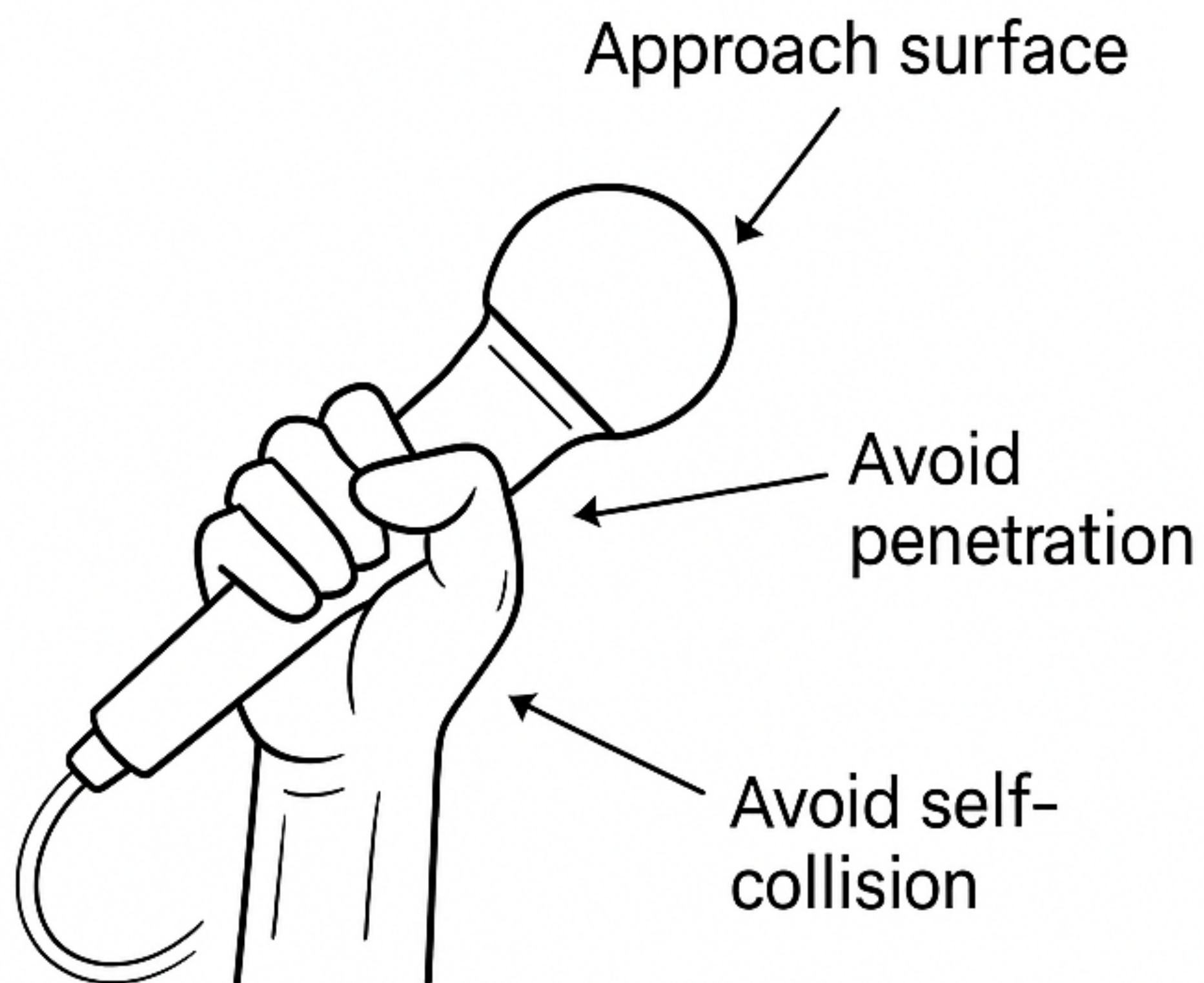
Physical violation



Unstable



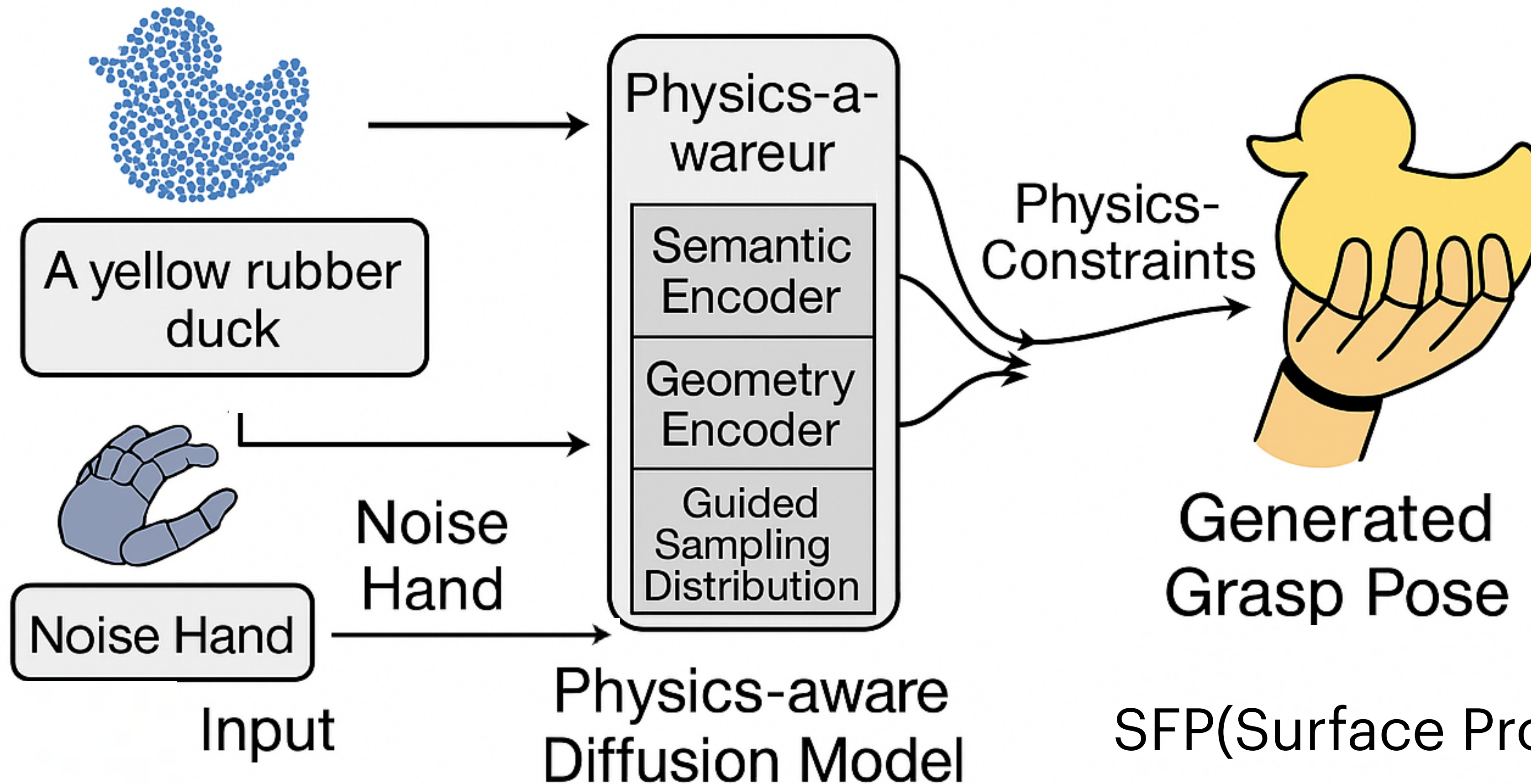
**✓** Physics-aware generation





# DexGrasp: Towards Universal Robotic Dexterous Grasping with Physics-Awareness

Sampling  
Process



SFP(Surface Proximity Force)

ERF(Elastic Repulsive Force)

SRF(Self-Repulsive Froce)



# Physics Constraints

## SPF

The contact must remain on the surface



$$L_{\text{SPF}} = \frac{\sum_{i \in S} \sqrt{d_i}}{|S| + \eta},$$

| Symbol | Meaning                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------|
| $d_i$  | Squared Euclidean distance between the $i^{\text{th}}$ hand point and its closest point on the object surface.   |
| $S$    | Set of hand surface points (e.g., contact candidates) that are within a certain threshold to the object surface. |
| $\eta$ | A small positive constant used to ensure numerical stability and avoid division by zero.                         |

## ERF

The object must not be penetrated



$$L_{\text{ERF}} = \frac{1}{B} \sum_{i=1}^B \max_i(s_i, d_i).$$

| Symbol             | Meaning                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| $B$                | Total number of sampled hand surface vertices used for collision checking.                                                |
| $d_i$              | Signed distance from the $i^{\text{th}}$ hand vertex to the object surface. Negative values indicate penetration.         |
| $s_i$              | A small positive constant or margin (e.g. 0 or slightly negative), representing a safe threshold.                         |
| $\max(s_i, d_i)$   | If $d_i < s_i$ , this term activates and contributes to the loss (i.e., when the hand is too close or inside the object). |
| $\frac{1}{B} \sum$ | Averaging the penalty across all hand vertices to ensure scale invariance.                                                |

## SRF

Collisions between fingers must be avoided



$$L_{\text{SRF}} = \frac{1}{B} \sum_{i,j} \max(0, d_{\text{Threshold}} - d_{ij}).$$

| Symbol                                   | Meaning                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $B$                                      | Total number of hand point pairs considered (usually depends on hand mesh size).                    |
| $d_{ij}$                                 | Euclidean distance between hand points $i$ and $j$ .                                                |
| $d_{\text{Threshold}}$                   | Minimum safe distance to avoid self-collision.                                                      |
| $\max(0, d_{\text{Threshold}} - d_{ij})$ | Penalty is applied only when the distance between points $i$ and $j$ is smaller than the threshold. |
| $\frac{1}{B} \sum$                       | Mean penalty over all considered point pairs.                                                       |



$$L_{\text{SPF}} = \frac{\sum_{i \in S} \sqrt{d_i}}{|S| + \eta},$$

| Symbol | Meaning                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------|
| $d_i$  | Squared Euclidean distance between the $i^{th}$ hand point and its closest point on the object surface.          |
| $S$    | Set of hand surface points (e.g., contact candidates) that are within a certain threshold to the object surface. |
| $\eta$ | A small positive constant used to ensure numerical stability and avoid division by zero.                         |

| Function          | Value at $d = 0.01$ | Value at $d = 100$ | Penalty Ratio (Far/Near)  |
|-------------------|---------------------|--------------------|---------------------------|
| $d^2$ (square)    | 0.0001              | 10,000             | 100,000× ⚠️ (too extreme) |
| $d$ (linear)      | 0.01                | 100                | 10,000×                   |
| $\sqrt{d}$ (sqrt) | 0.1                 | 10                 | 100× ✅ (balanced)         |

The purpose of using sqrt(di) is to impose a mild penalty on a distant point. This makes the loss smoother and the training more stable,



# Physics Constraints

## SPF

The contact must remain on the surface



$$L_{\text{SPF}} = \frac{\sum_{i \in S} \sqrt{d_i}}{|S| + \eta},$$

| Symbol | Meaning                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------|
| $d_i$  | Squared Euclidean distance between the $i^{\text{th}}$ hand point and its closest point on the object surface.   |
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## ERF

The object must not be penetrated



$$L_{\text{ERF}} = \frac{1}{B} \sum_{i=1}^B \max_i(s_i, d_i).$$

| Symbol             | Meaning                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| $B$                | Total number of sampled hand surface vertices used for collision checking.                                                |
| $d_i$              | Signed distance from the $i^{\text{th}}$ hand vertex to the object surface. Negative values indicate penetration.         |
| $s_i$              | A small positive constant or margin (e.g. 0 or slightly negative), representing a safe threshold.                         |
| $\max(s_i, d_i)$   | If $d_i < s_i$ , this term activates and contributes to the loss (i.e., when the hand is too close or inside the object). |
| $\frac{1}{B} \sum$ | Averaging the penalty across all hand vertices to ensure scale invariance.                                                |

## SRF

Collisions between fingers must be avoided



$$L_{\text{SRF}} = \frac{1}{B} \sum_{i,j} \max(0, d_{\text{Threshold}} - d_{ij}).$$

| Symbol                                   | Meaning                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $B$                                      | Total number of hand point pairs considered (usually depends on hand mesh size).                    |
| $d_{ij}$                                 | Euclidean distance between hand points $i$ and $j$ .                                                |
| $d_{\text{Threshold}}$                   | Minimum safe distance to avoid self-collision.                                                      |
| $\max(0, d_{\text{Threshold}} - d_{ij})$ | Penalty is applied only when the distance between points $i$ and $j$ is smaller than the threshold. |
| $\frac{1}{B} \sum$                       | Mean penalty over all considered point pairs.                                                       |



$$L_{\text{ERF}} = \frac{1}{B} \sum_{i=1}^B \max_i(s_i, d_i).$$

| Symbol             | Meaning                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| $B$                | Total number of sampled hand surface vertices used for collision checking.                                                |
| $d_i$              | Signed distance from the $i^{th}$ hand vertex to the object surface. Negative values indicate penetration.                |
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| $\max(s_i, d_i)$   | If $d_i < s_i$ , this term activates and contributes to the loss (i.e., when the hand is too close or inside the object). |
| $\frac{1}{B} \sum$ | Averaging the penalty across all hand vertices to ensure scale invariance.                                                |

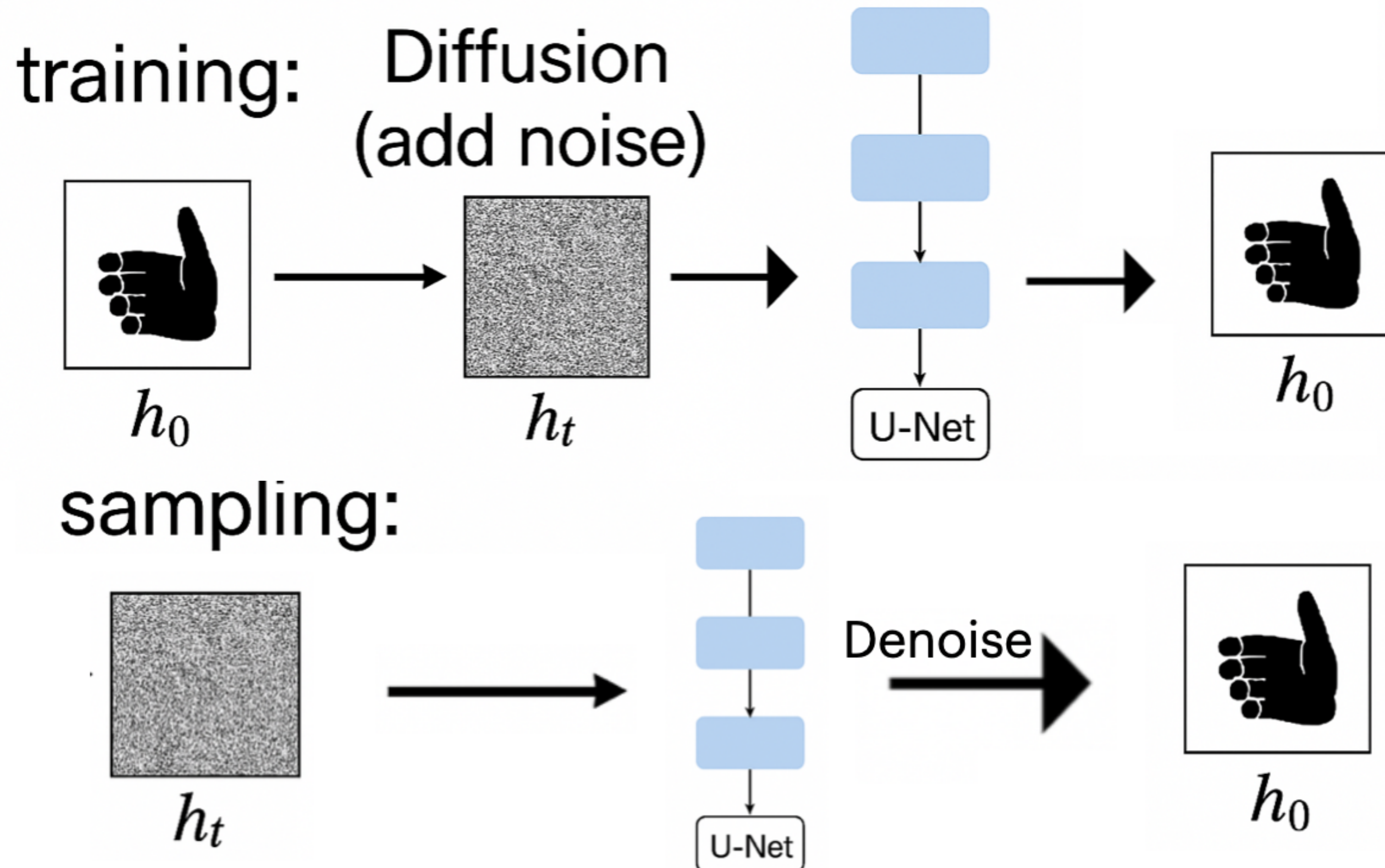
$$L_{\text{SRF}} = \frac{1}{B} \sum_{i,j} \max(0, d_{\text{Threshold}} - d_{ij}).$$

| Symbol                                   | Meaning                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $B$                                      | Total number of hand point pairs considered (usually depends on hand mesh size).                    |
| $d_{ij}$                                 | Euclidean distance between hand points $i$ and $j$ .                                                |
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| $\max(0, d_{\text{Threshold}} - d_{ij})$ | Penalty is applied only when the distance between points $i$ and $j$ is smaller than the threshold. |
| $\frac{1}{B} \sum$                       | Mean penalty over all considered point pairs.                                                       |

The ERF ensure that the entire surface of the hand does not penetrate into the interior of the target object. The SRF guides the fingers to maintain a reasonable spacing to avoid physical irrationalities.

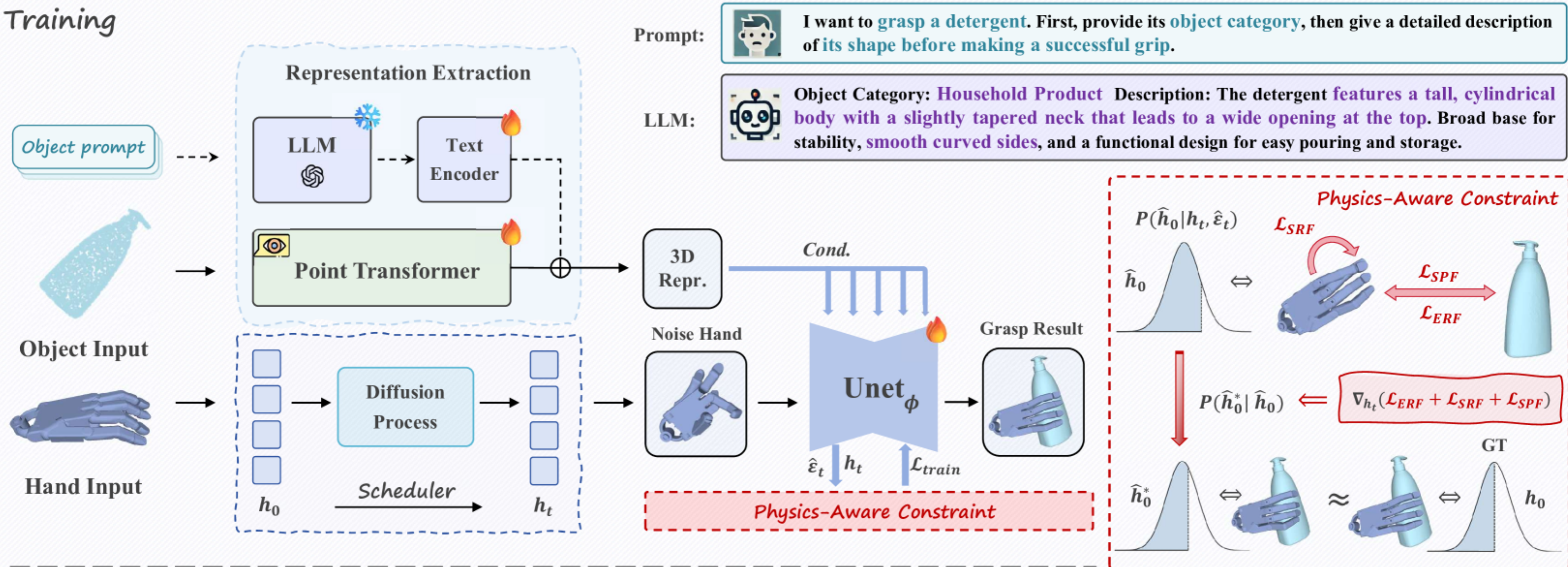


# Defusion model

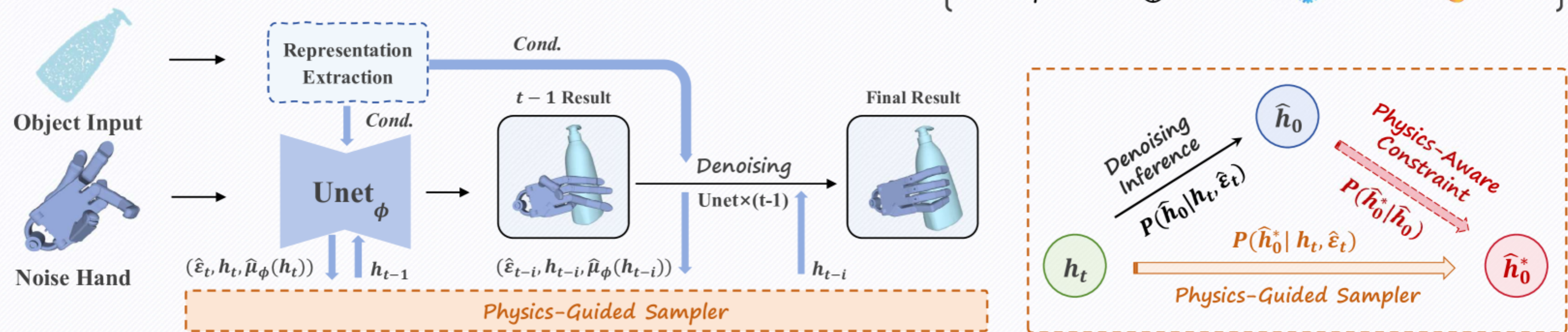




## Training



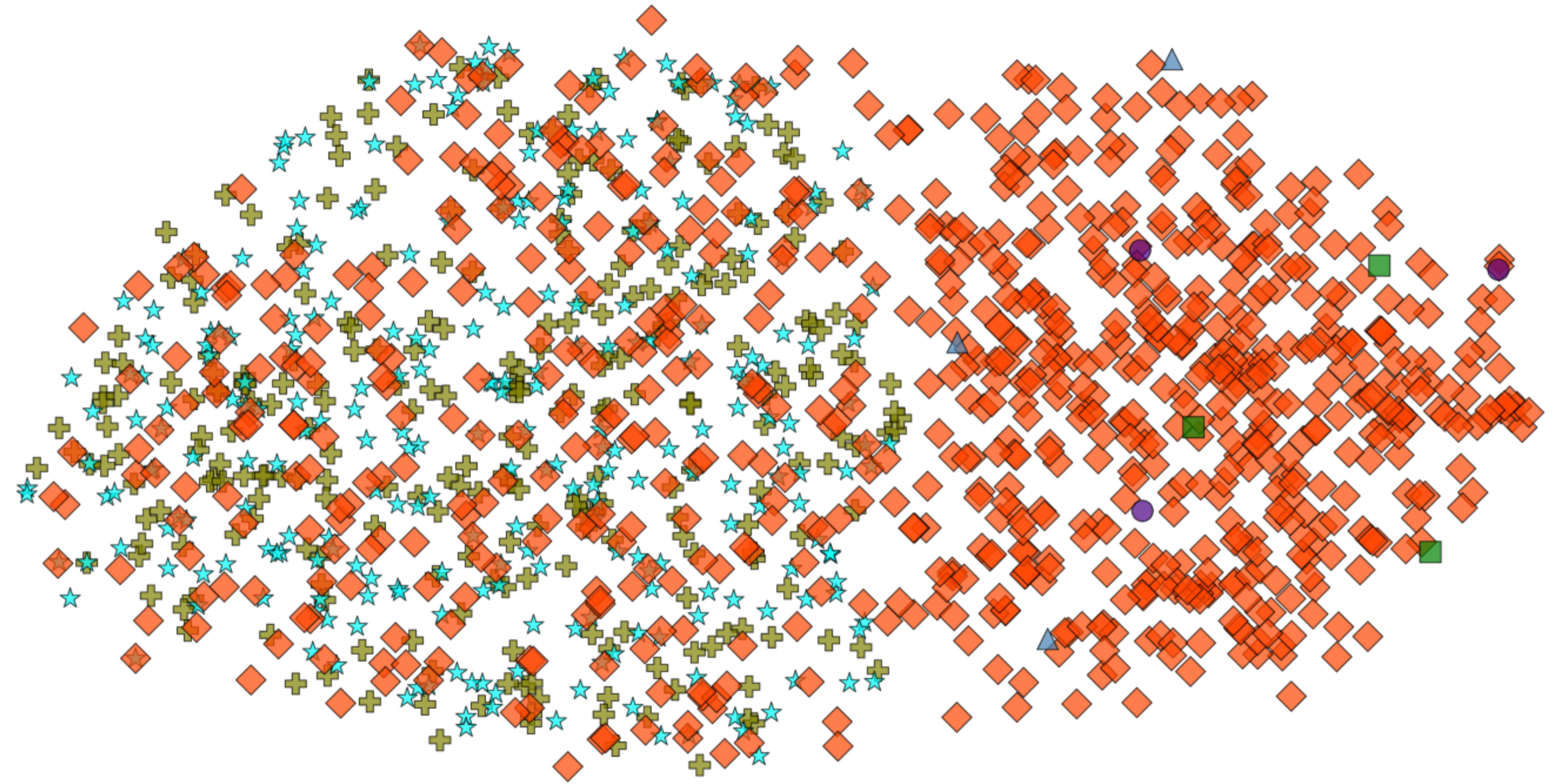
## Sampling





# DexGrasp Big Datasets

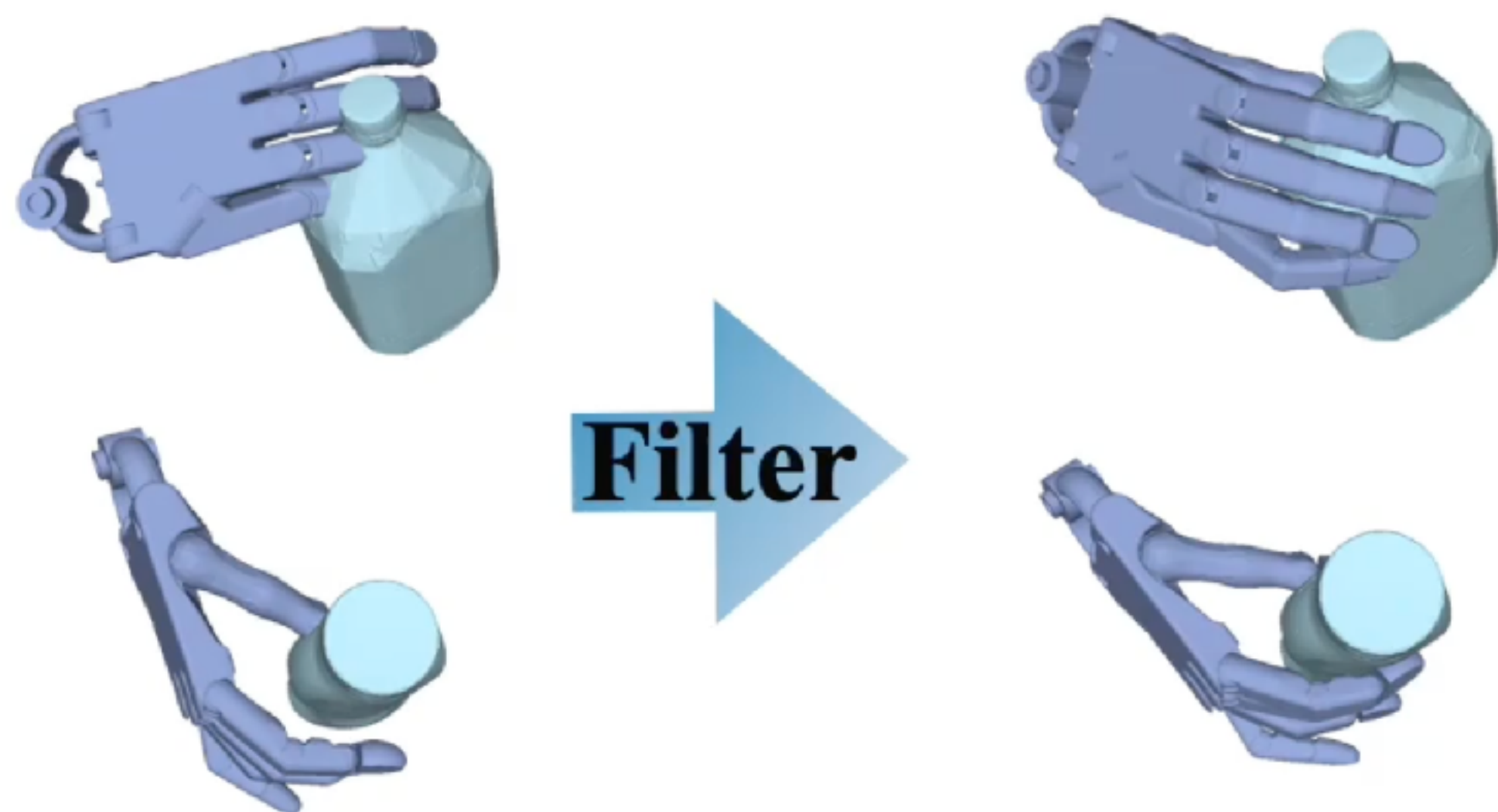
| Dataset           | # Grasps  |
|-------------------|-----------|
| DGA-curated       | 0.88M     |
| DGA-generated     | 2.52M     |
| Total DGA         | 3.40M+    |
| DexYCB            | ~0.37M    |
| ContactDB         | ~0.27M    |
| GraspNet-1Billion | ~0.97M (c |



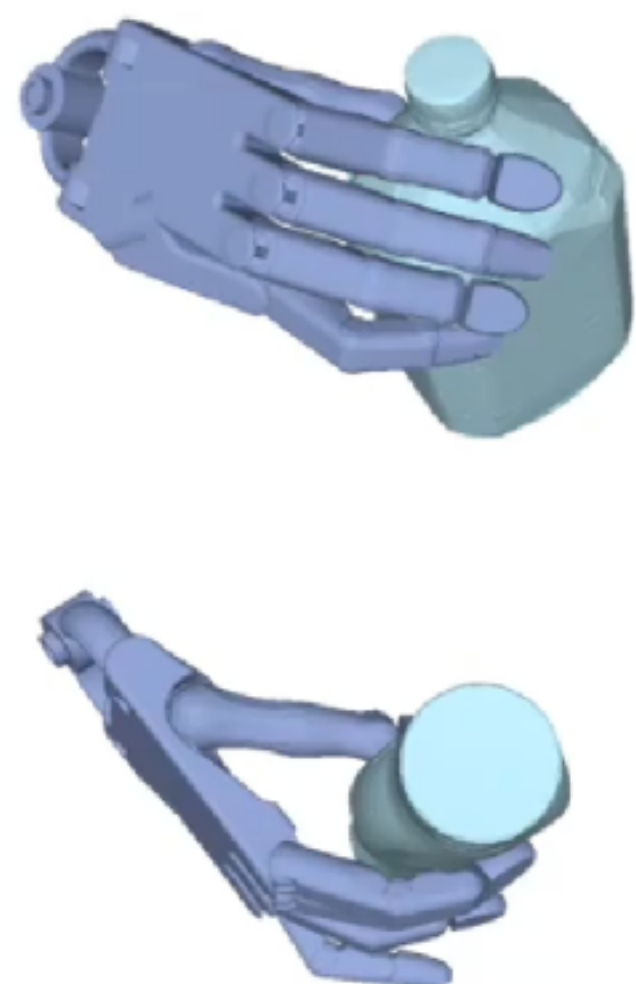
● **MultiDex** ■ **RealDex** ▲ **DexGRAB** + **UniDex** ★ **Dexgraspnet** ◆ **Ours**



# Curation and Filtering

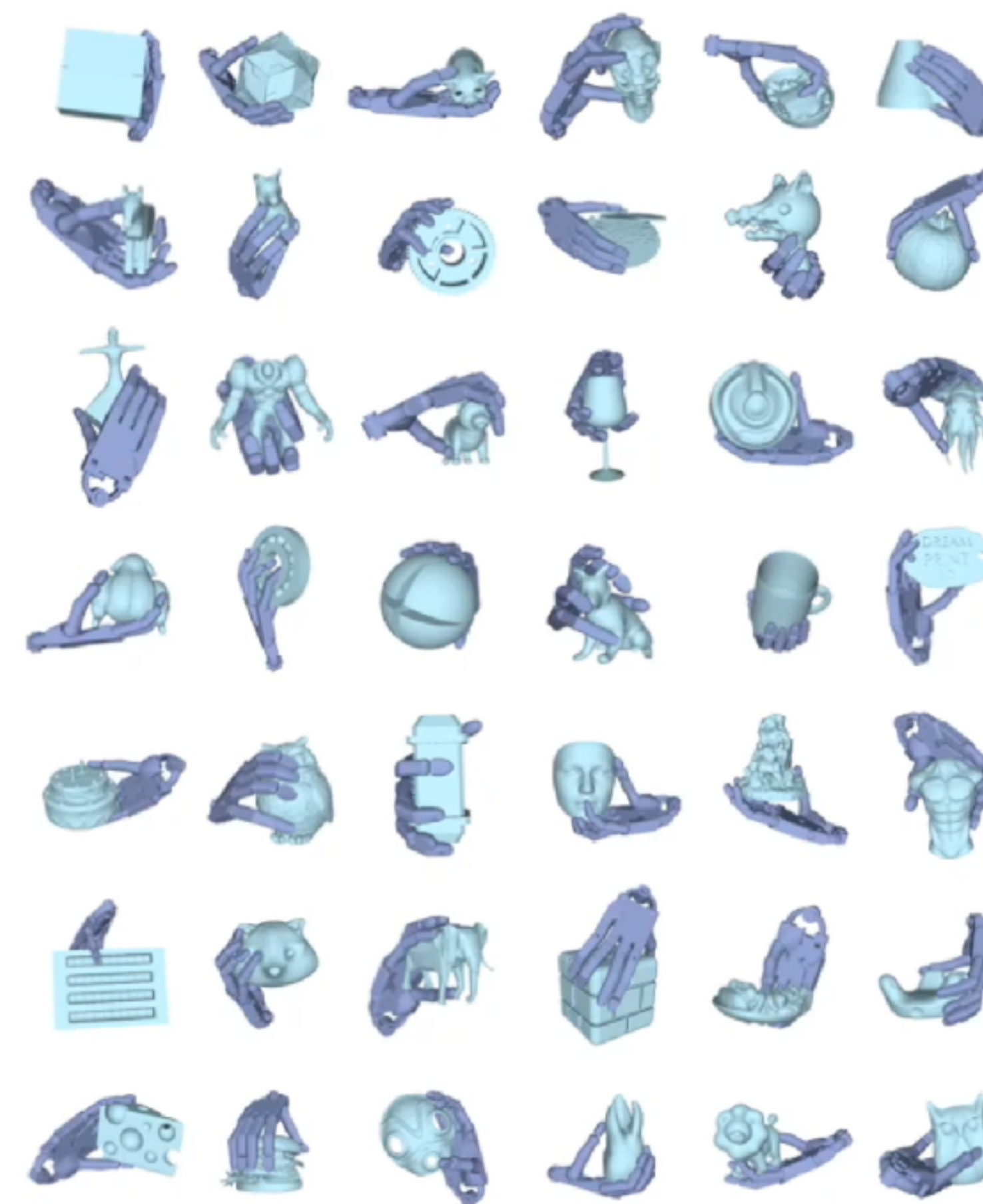


**Filter**



**zero-shot**

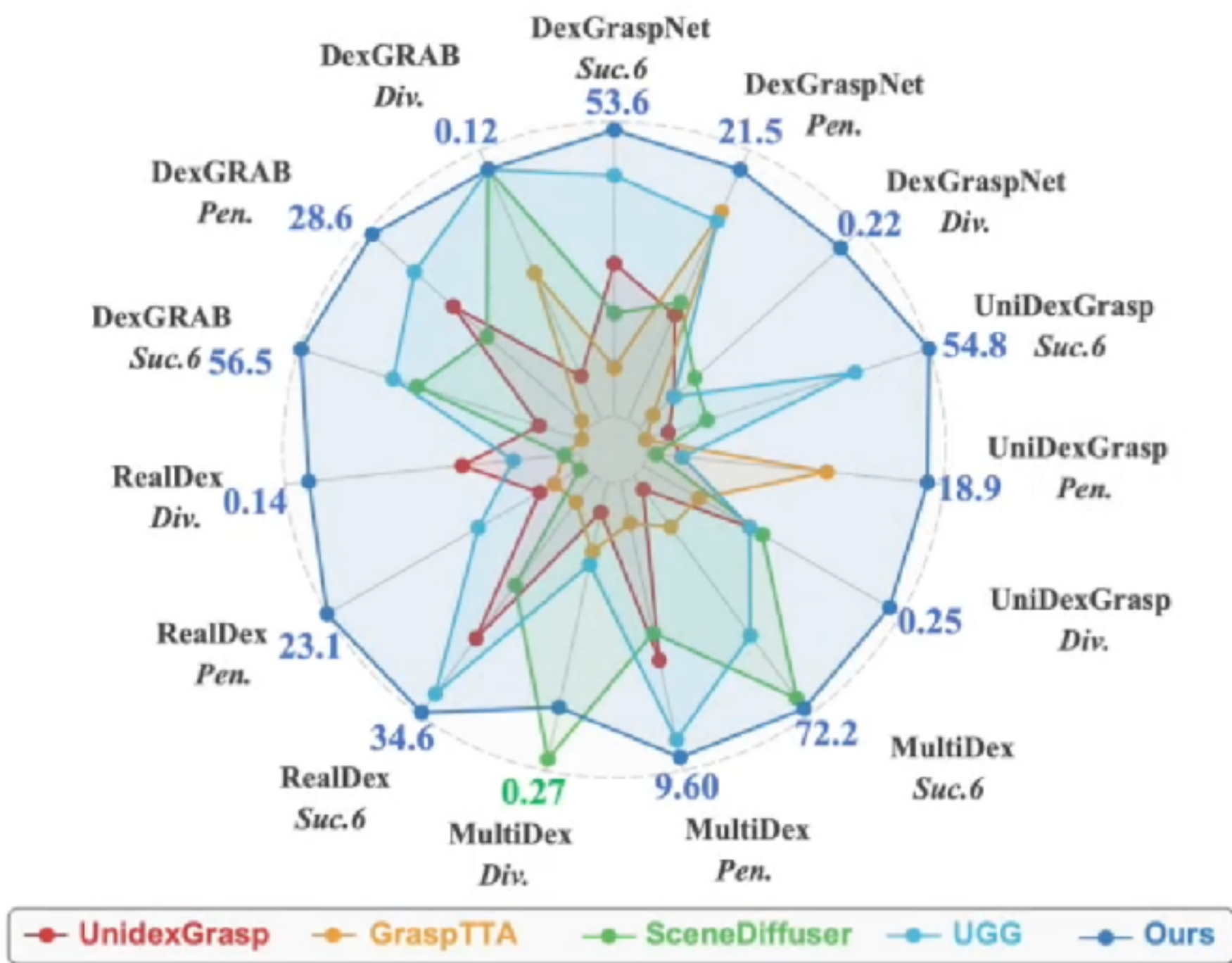
**model-in-the-loop**



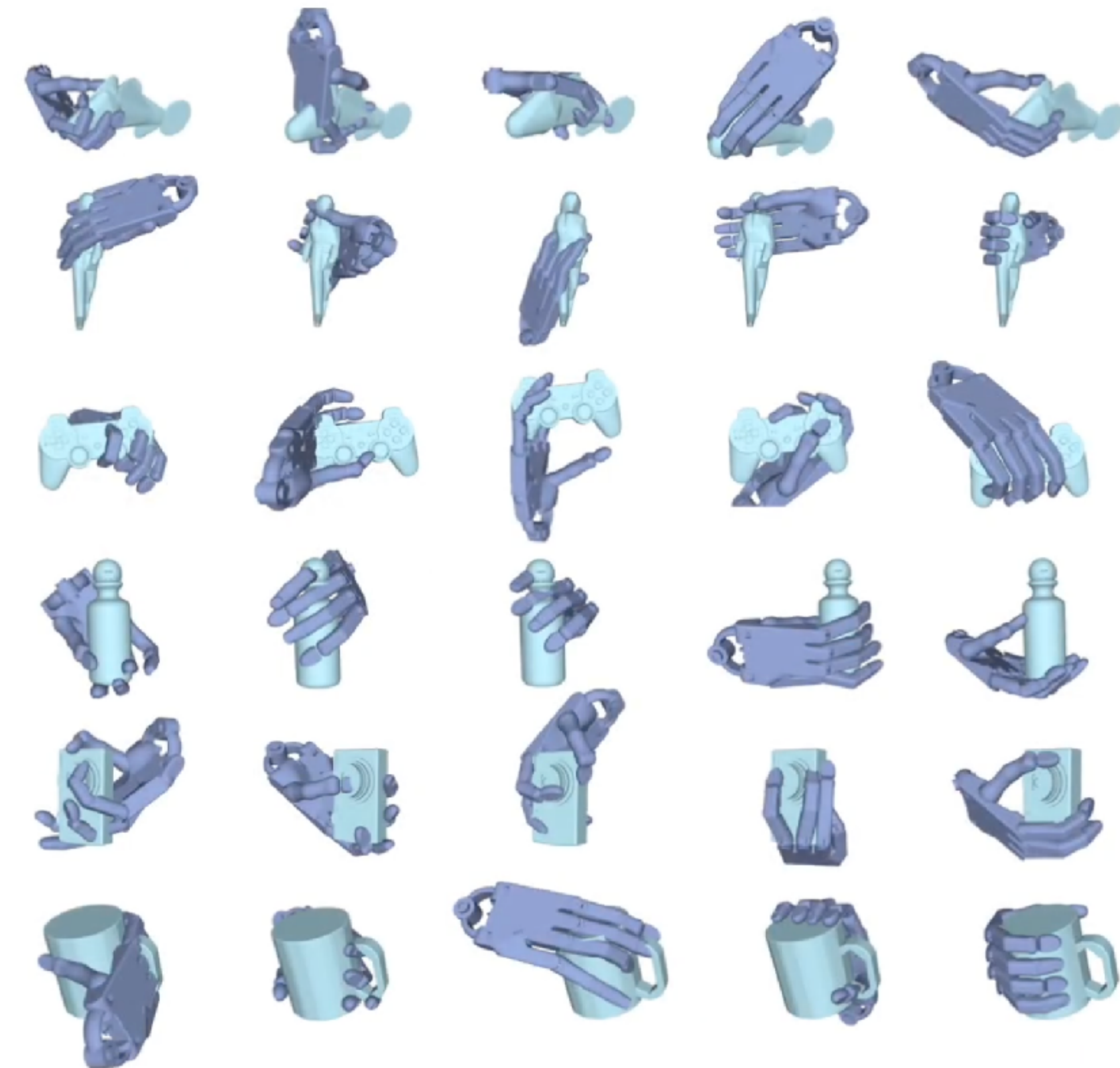
**Object mesh more complex more smooth and more diverse**



# Comparison

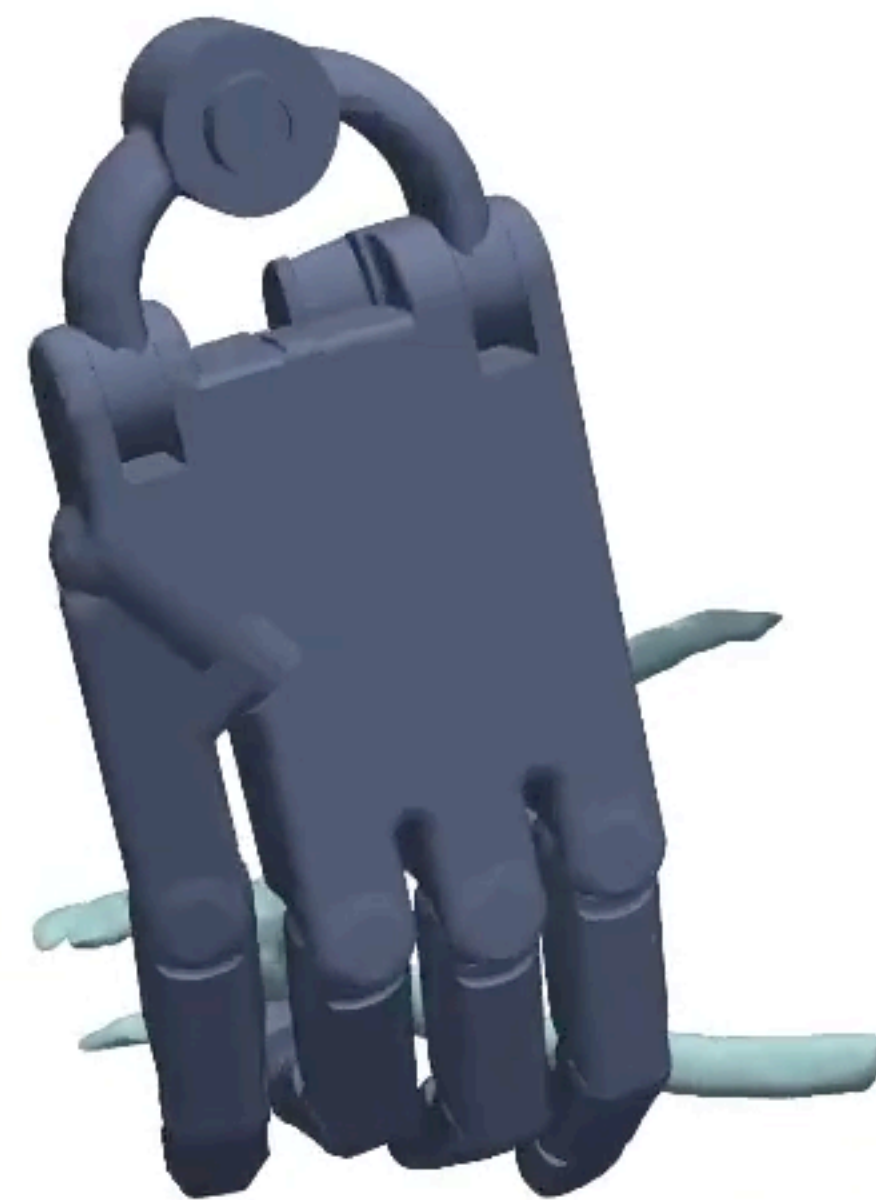


Suc.6 → Success rate (6 directions) Pen. → Penetration Div. → Diversity

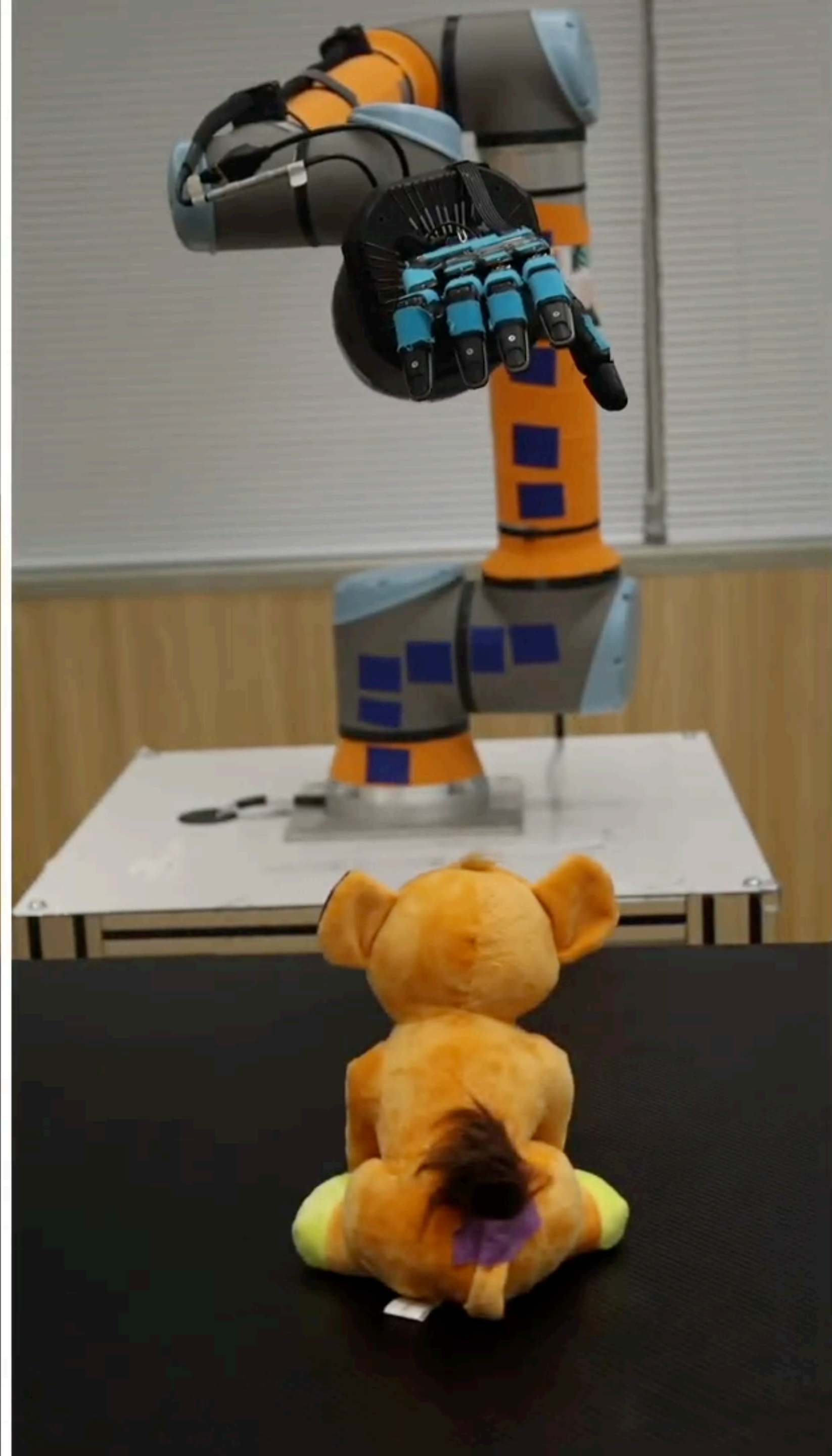
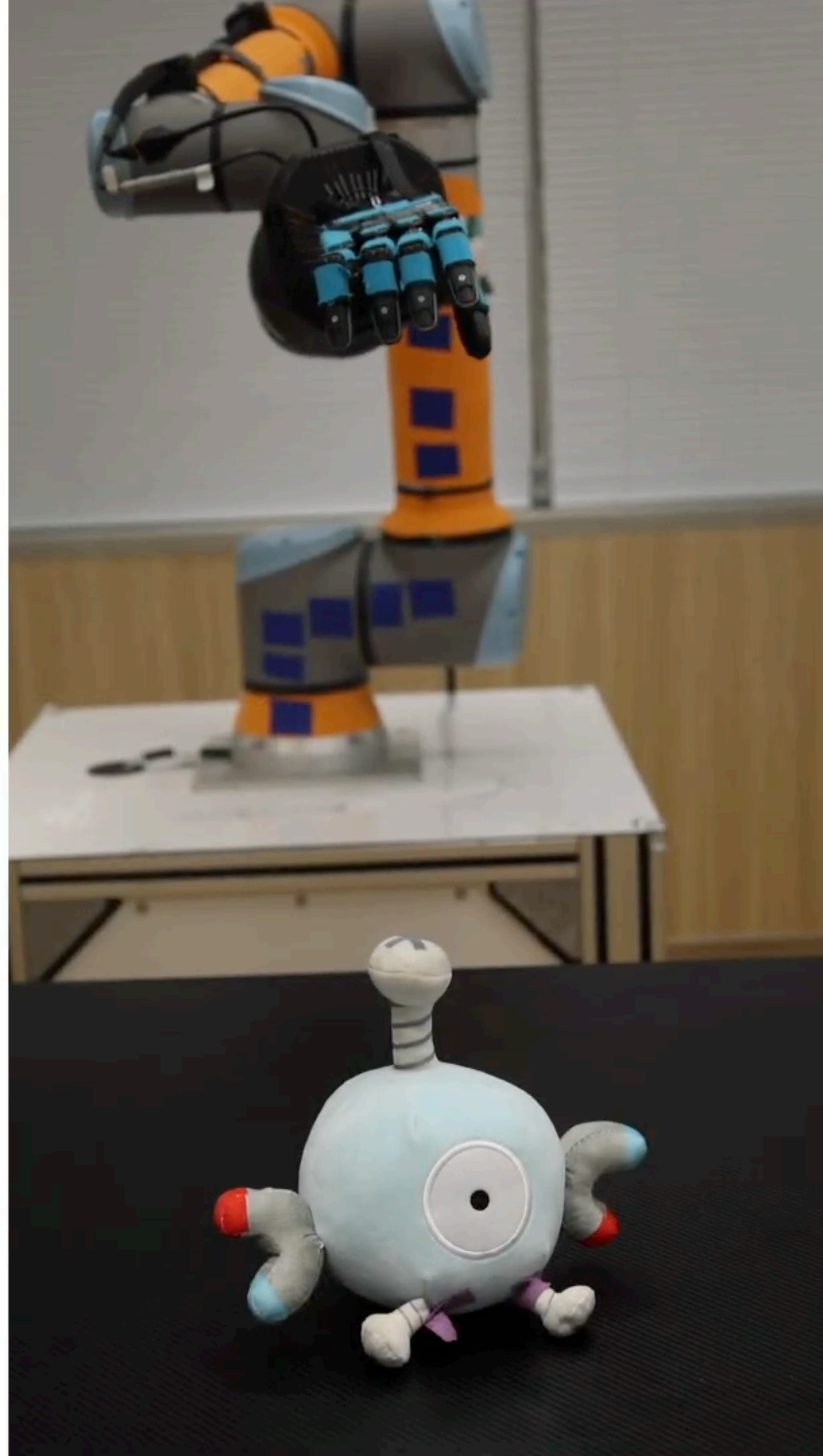
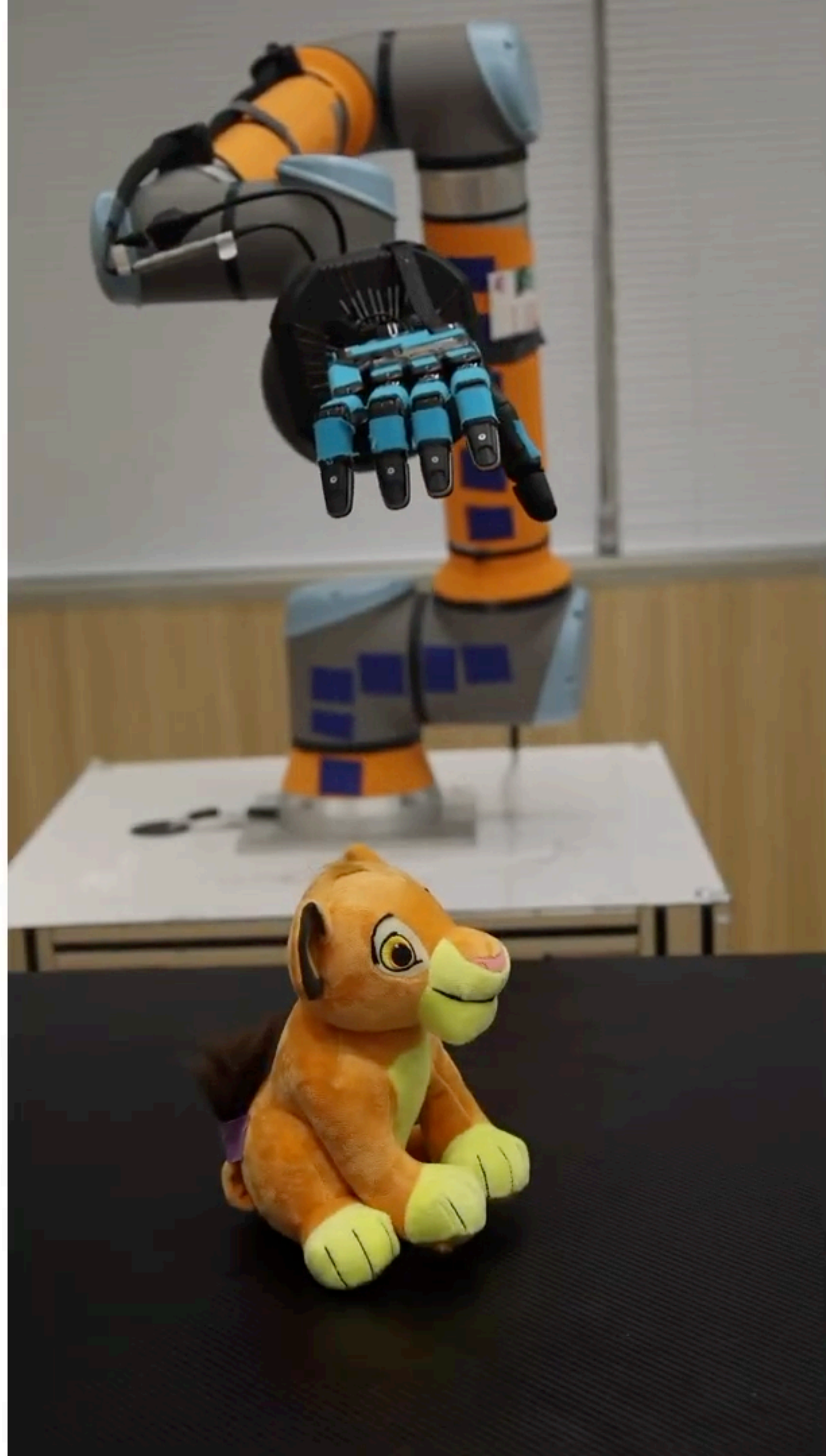


UniDexGrasp GraspTTA SceneDiffuser UGG Ours











# Q&A